Thomas Coen

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EDUCATION

Duke University, Pratt School of Engineering **Durham, NC**

B.S.E. Mechanical Engineering, Class of 2029 · Aerospace Engineering Certificate *GPA: 3.78*

Relevant coursework: Thermodynamics, Mechanics of Solids, Differential Equations, Engineering Design

ENGINEERING EXPERIENCE

Duke AERO (Rocketry Team) **Durham, NC**

Structures Lead *July 2026 – Present*

- Lead structural design for a 30,000 ft student researched and developed (SRAD) rocket for the 2027 International Rocket Engineering Competition (IREC) and a separate two-stage staging test rocket
- Tune and troubleshoot the team's filament winder for future fiberglass and carbon tubes; completed a test wind (Oct. 2026). Target faster, cheaper, potentially stronger tubes than hand layup; stress-strain testing is pending

Structures & Solid Propulsion Member *2025 – 2026*

- Machined a 6061 aluminum nozzle casing on a CNC lathe; hardware survived full-duration static hot fire and flight. The team placed 3rd in the 30,000 ft SRAD propulsion category at IREC 2026
- Cut prepreg carbon fiber and helped lay up rocket body tubes
- Machined canard and air brake housing components on a Viwa CNC mill
- Assembled motor hardware for static hot fire and flight
- Supported operations at IREC 2026 in Midland, TX, and helped recover the rocket after flight

BreakerBots, FIRST Robotics Competition Team 5104 **Pacific Grove, CA**

Co-Captain, CAD & Manufacturing Lead (2025) *2022 – 2025*

- Redesigned Leviathan's 3-stage cascade elevator (9.5 ft travel, 2× Kraken X60 motors, 1:5 reduction) with constant force spring assist, cutting extension time from 3.5 s to 1.8 s for an estimated 25% more scoring cycles (2025)
- Led a robot-wide weight reduction from 134 lb to 114 lb (15%), below the 115 lb limit. Pocketed aluminum elevator structure and replaced aluminum brackets with 3D printed carbon fiber (2025)
- Coordinated 5 subteams and 35 members to build Leviathan, the team's most technically complex robot (2025)
- Designed and machined Typhoon's intake pivot and intake, balancing packaging, backlash, and weight. Created CAD models and CAM toolpaths, machined structural parts, and grew the manufacturing subteam to 15+ machinists (2024)
- As Manufacturing Lead, grew the subteam from 2 to 8+ machinists and created CAM toolpaths and machined structural components for Kraken and Cthulhu (2023)
- Machined aluminum tubing, routed the bellypan, and fabricated polycarbonate panels for Charybdis as a first-year machinist (2022)
- Progressed from machinist (2022) to Manufacturing Lead (2023), Co-Captain (2024), and CAD Lead (2025); served as Drive Coach from 2023 to 2025

Duke EGR 121 — Sunflower Seed Sheller **Durham, NC**

Mechanical Design *2026*

- Designed the rollers, drive mechanism, and funnel; created all CAD models for a 4-person team
- Iterated roller geometry to crack shells along the seam while limiting kernel damage
- Team produced 41 intact kernels in 3 minutes, earning the top section score (8 teams) on yield minus waste and 2nd among all first-year mechanical engineering teams

SKILLS

Design: SolidWorks, Onshape, Fusion 360 (CAD/CAM), AutoCAD

Manufacturing: CNC milling (Tormach 1100MX, 770M, Viwa), CNC turning, manual lathe, 3D printing, carbon fiber and fiberglass layup, filament winding, laser cutting; Titans of CNC certification series